MIL-B-51298A(MU)
18 February 1971
SUPERSEDING
MIL-B-51298(MU)
11 October 1968

MILITARY SPECIFICATION

BODY ASSEMBLY, M674 and M675

CARTRIDGES

1. SCOPE

1.1 This specification covers one type of virgin butyl rubber body for a cartridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Fiberboard.

MILITARY

MIL-C-60303 - Cartridge, 40MM, Riot Control, CS, M674.

STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.
MIL-STD-1168 - Lot Numbering of Ammunition.

FSC 1310

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

C122-3-104 - Body Assembly.
B143-14-7 - Riot Mixture, CS.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D1149-60T - Test for Accelerated Ozone Cracking of Vulcanized Rubber.
D736-54T - Test for Low-Temperature Brittleness of Rubber and Rubber-Like Materials.

(Application for copies of the above publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street, N.W., Washington, D.C. 20036.)

Agent, T.C. George's Tariff

(Application for copies should be addressed to T.C. George, Agent, 2 Penn Plaza, New York, N.Y. 10001.)

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3. REQUIREMENTS

3.1 Material and components.

3.1.1 Material. The body assembly shall be made of virgin butyl synthetic rubber in accordance with Drawing C122-3-104 (see 6.3).

3.1.2 Components. All components of the body assembly shall conform to Drawing C122-3-104.

3.2 Manufacture and assembly. The components shall be manufactured and assembled as specified in Drawing C122-3-104.

3.3 Port membrane and body leakage. The four port membranes in the body shall not leak when subjected to a minimum air pressure of 1 pound per square inch gage (psig) and tested as specified in 4.4.4.1. The body shall not leak when subjected to an air pressure of 6 to 10 psig and tested as specified in 4.4.4.1.

3.4 Physical properties. Physical properties of the material for the body shall conform to the requirements in table I when tested as specified therein.

Table I. Physical properties of body assembly

Property	As received	Aged, 168 hours at 158° ± 1.8° Fahrenheit	Test paragraph
200% modulus psi	200 to 500	200 to 500	4.4.4.2(a)
Tensile strength, psi	900 min	775 min	4.4.4.2(b)
Elongation, %	500 min	25% increase 30% decrease	4.4.4.2(c)
Hardness, shore A	65 ± 5	65 (+10, -5)	4.4.4.2(d)
Tear strength, lb/inch thick	100	75	4.4.4.2(e)
Body weight, grams	88 ± 3	88 ± 3	4.4.4.4
Tension set %	25 max		4.4.4.2(i)
Ozone resistance	No cracking after exposure to 25 ± 5 *pphm at 100°F for 48 hours		4.4.4.2(f)
Flammability	Self extinguishing within 10 sec		4.4.4.2(g)
Low temperature	No cracks or shatter after blow, following exposure to -15°F for five hours		4.4.4.2(h)

*pphm - parts per hundred million.

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3.5 Permeability to CS. The loss of CS shall not exceed 4.65 percent when tested as specified in 4.3.3. Once the rubber formulation has been approved by the Government, the supplier shall not make any change in the formulation or in the curing process without obtaining Government approval.

3.6 Port membrane. The thickness of the port membrane, the diameter of the flat of the membrane, and the radius of the flat to the side wall shall be in accordance with Drawing C122-3-104 when tested as specified in 4.4.4.3.

3.7 Functioning. When specified in the contract or order, the body shall be filled with CS, assembled into cartridges and tested as specified in 4.4.4.5 (see 6.2(c)).

3.8 Preproduction. Prior to the start of regular production, a preproduction sample of bodies shall be produced in accordance with this specification for examination and test (see 4.3).

3.9 Workmanship. The body shall be free from imperfections listed in table I of MIL-STD-407 and damage such as rips, tears, or punctures which could affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.1.3 Permeability to CS. Within 30 days after the date of award of a contract, and without additional cost to the Government, the contractor shall submit the information listed below which will constitute his evidence for establishing the permeability to CS of the rubber formulation.

4.1.3.1 Information relating to candidate formulation.

(a) The name of the contractor and the producing subcontractor, if any.

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(b) A description of the candidate formulation that will include:

(1) The chemical and trade names of the ingredients.

(2) The quantity of each ingredient (parts by weight), including the range of variation that can be tolerated without loss of quality in the finished product.

(3) The mold lubricants, dusting powders, or wet sprays proposed.

(4) Proposed cure times and temperature.

(5) Results of previous permeability to CS evaluation demonstrating acceptability or evidence of prior Government approval of the proposed formulation.

4.1.3.2 Testing. If the above information is not sufficient to establish permeability to CS of the butyl rubber formulation, the contractor shall furnish the Government with 4 bodies from each mold cavity for performing the test specified in 4.3.3.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 21 bodies from each mold cavity shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. Each body in the preproduction sample shall be examined in accordance with the classification of defects (4.4.3.3). The AQL does not apply.

4.3.2.2 For tests. The sample bodies manufactured as specified in 4.3.1 shall be tested in accordance with table II.

4.3.2.3 By the Government. One of the four bodies selected to be inspected in accordance with 4.4.4.3(a) (see table II) shall be forwarded for inspection at a Government facility (see 6.2(b)).

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4.3.3 Test for permeability to CS. The bodies when filled with CS riot mix conforming to Drawing B143-14-7, shall be placed in a hot air oven for 90 days at 168° ± 5°F. The bodies shall be loaded with CS riot mix only. The delay mixture, first fire mixture, and quickmatch shall be omitted. A suitable closure seal shall be used at neck of body.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examination and tests specified in 4.3.2 to be acceptable The acceptance number is zero. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

Table II. Preproduction sampling and test

*Mold cavity quantity	Requirement	Examination or test
16 from each cavity	3.3 Leakage	4.4.4.1
16 from each cavity	Dwg C122-3-104	4.4.3.3
4 from each cavity	table I	4.4.4.2(a) to 4.4.4.2(i)
2 from each cavity	3.5 Permeability	4.3.3
4 from each cavity	3.6 Port membrane	4.4.4.3(a)
4 from each cavity	table I	4.4.4.4
5 from each cavity	3.7 Functioning	4.4.4.5

*189 bodies from nine - cavity mold.
677 bodies from 37 - cavity mold.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. However, no more than one week's production shall be represented in any one lot of bodies.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. Approximately equal number of bodies shall be selected from each cavity.

4.4.2.2 For destructive test. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

4.4.2.3 For inspection by the Government. There shall be taken from each lot a sample of one body from each cavity from one mold (the mold from which the sample is drawn will be alternated with each lot if two molds are used in

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producing the lot) for inspection in accordance with 4.4.4.3(b) at a Government facility. In lieu of whole bodies, upper portions of bodies containing all four ports with membranes intact may be submitted for this inspection.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The sample bodies shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive test. The sample bodies shall be tested in accordance with 4.4.4.2, using an AQL of 1.0 percent defective.

4.4.3.3 Classification of defects.

(a) Body assembly (Dwg C122-3-104).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 2.5 percent defective	
101	Leakage	4.4.4.1
102	Length of neck incorrect	
103	Inside diameter of neck incorrect	
104	Wall thickness of neck incirrect	
105	Length of body incorrect	
106	Outside diameter of body incorrect	
*107	Wall thickness of body incorrect	
108	Workmanship (see 3.9)	

*To be checked prior to vulcanizing end in body.

4.4.4 Tests.

4.4.4.1 Port membrane and body leakage. The body shall be subjected to the specified pressure using a suitable adapter and submerged in water at ambient temperature, to a depth of 2 to 3 inches over the body for a period of 30 $\pm$ 5 seconds. The water shall contain a compatable wetting agent. Evidence of steady bubbles shall indicate leakage.

4.4.4.2 Physical properties. The body shall be cut circumferentially into approximately three strips after removing the two ends. The neck end shall be used to check port membrane thickness, diameter of flat portion of membrane and the radius of membrane to side wall. The strips shall be tested as specified in 4.4.4.2(a) to 4.4.4.2(i).

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(a) Two hundred percent modulus psi. The test specimen shall be tested in accordance with method 4131 of Federal Test Method Standard Number 601.

(b) Tensile strength. The test specimen shall be tested in accordance with method 4111 of Federal Test Method Standard Number 601.

(c) Elongation. The test specimen shall be tested in accordance with method 4121 of Federal Test Method Standard Number 601.

(d) Hardness, Shore A. The test specimen shall be tested in accordance with method 3021 of Federal Test Method Standard Number 601.

(e) Tear strength, die C. The test specimen shall be tested in accordance with method 4211 of Federal Test Method Standard Number 601.

(f) Ozone resistance. The test specimen shall be tested in accordance with ASTM D1149-60T.

(g) Flammability. The tip of the test specimen shall be held horizontally at the apex of the blue inner flame of a bunsen burner for 15 seconds. The total height of the flame shall be approximately 2-1/2 inches. The test shall be conducted in a draft-free enclosure. After the above specified time, the specimen shall be removed from the flame and while continuing to be held at the horizontal, the time in seconds for the flame to self-extinguish shall be measured.

(h) Low temperature. The test specimen shall be tested in accordance with ASTM D736-54T.

(i) Tension set. The test specimen shall be tested in accordance with method 4411 of Federal Test Method Standard Number 601.

4.4.4.3 Port membrane.

(a) Preproduction. The thickness of the port membrane shall be measured. The diameter of the flat of the membrane and the radius of the fillet at the junction of the membrane and side wall shall be verified by inspecting a cross-section cut, with the aid of a tool-maker's microscope and shadowgraph on the sample body.

(b) Regular production. The thickness of the port membrane shall be measured. This measurement, if satisfactory, and a satisfactory result in the test of 4.4.4.1, combined with the fact that preproduction sample approval has been granted, shall be accepted as proof that the port membrane area dimensions are satisfactory.

4.4.4.4 Body weight. Body weight shall be determined using a suitable gram scale with an accuracy of 0.5 gram.

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4.4.4.5 Functioning. The assembled cartridges shall be functioned in accordance with MIL-C-60303 for evaluation by a Government technical team.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing (interplant shipment). A designated quantity of items, not to exceed the weight limitations specified as follows, shall be packed in containers or special packs as applicable together with interior packaging, constructed as to give assurance the items will remain in prime condition and adequate to assure safe delivery to destination without damage to contents. Containers shall comply with Uniform Freight Classification, National Motor Freight Classification or Agent T.C. George's Tariff, as applicable, and be acceptable for shipment at the most favorable rate. The containers shall be sufficiciently rigid to withstand transportation and permit storing in tiers at least ten feet high without damage to containers or contents. The gross weight and size limitations of the containers shall not exceed the tariff or classification limitations specified therein, and the containers shall be closed and strapped as specified therein (see 6.4).

5.2 Marking. In conjunction with information provided in the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The rubber body is a component of the M674 and M675 cartridges. It is intended for containing the CS fill or red smoke mixture.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instruction when testing is performed by the Government.
(3) Time required for the Government to notify the applicator whether or not to proceed with production.

(c) When bodies are to be filled and assembled into cartridges.

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6.3 Formula for rubber compound. The following formula has been found to produce satisfactory bodies:

	Parts by weight
Butyl 268	100.0
Zinc oxide	5.0
FEF black	40.0
Stearic acid	1.0
Oncar 23-A	20.0
Sulfur	1.0
Tuex	2.0
Sulfasan R	1.5
MBT	0.5
Dechlorane	40.0
Retarder PB	1.5
	212.5

Use of this formula does not relieve the contractor from meeting the requirements of this specification.

6.4 Interplant shipment. Material which will not enter the military supply system. Typical are shipments of component parts from a vendor to a prime contractor.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1310-A131

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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